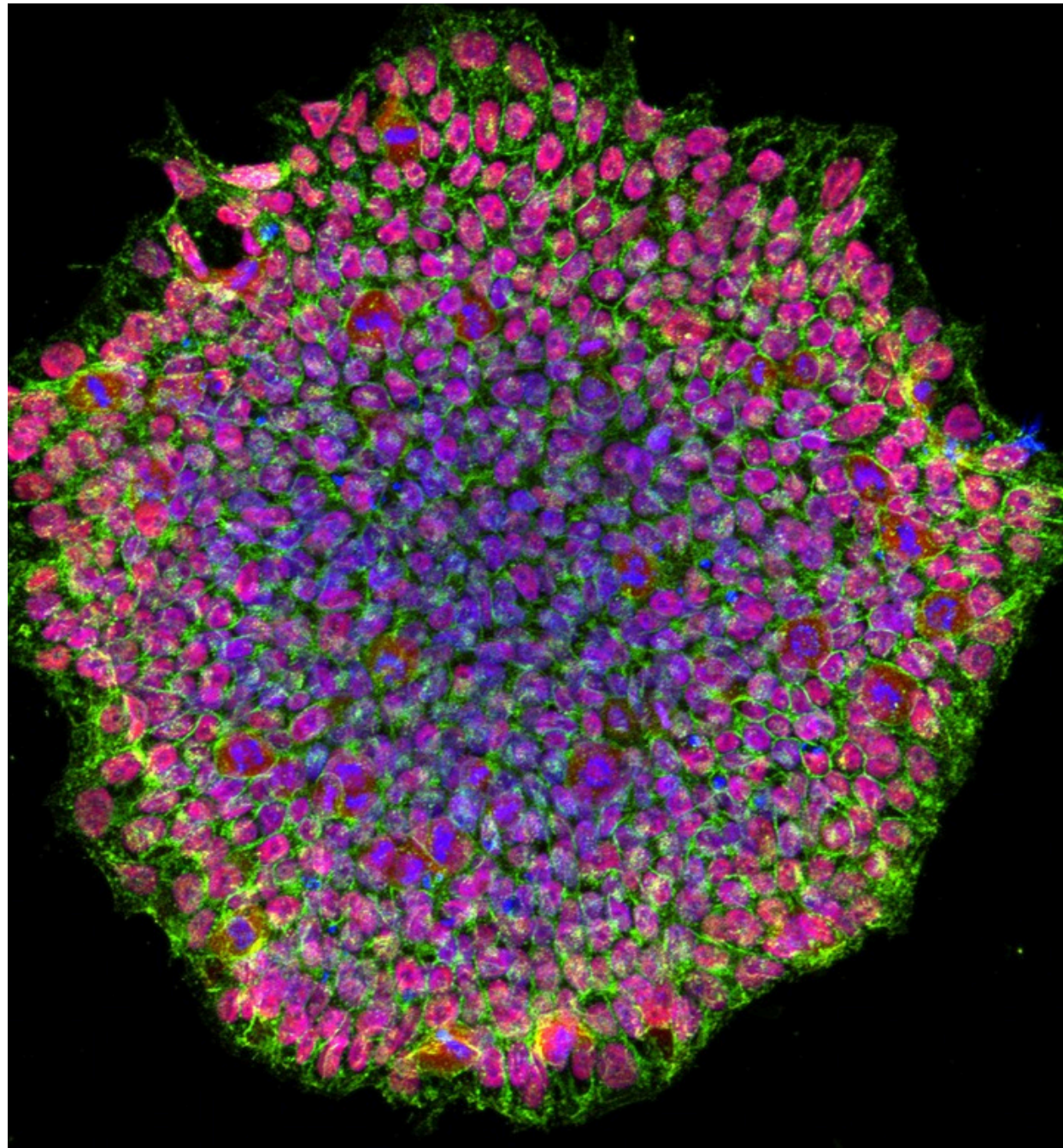


FOXA2 S169P Variant: Structural and Functional Implications in iPSC Differentiation

Student Name: ____宋子淇____ **Research Advisor:** ____耿老师____

The iPSC technology and FOXA2



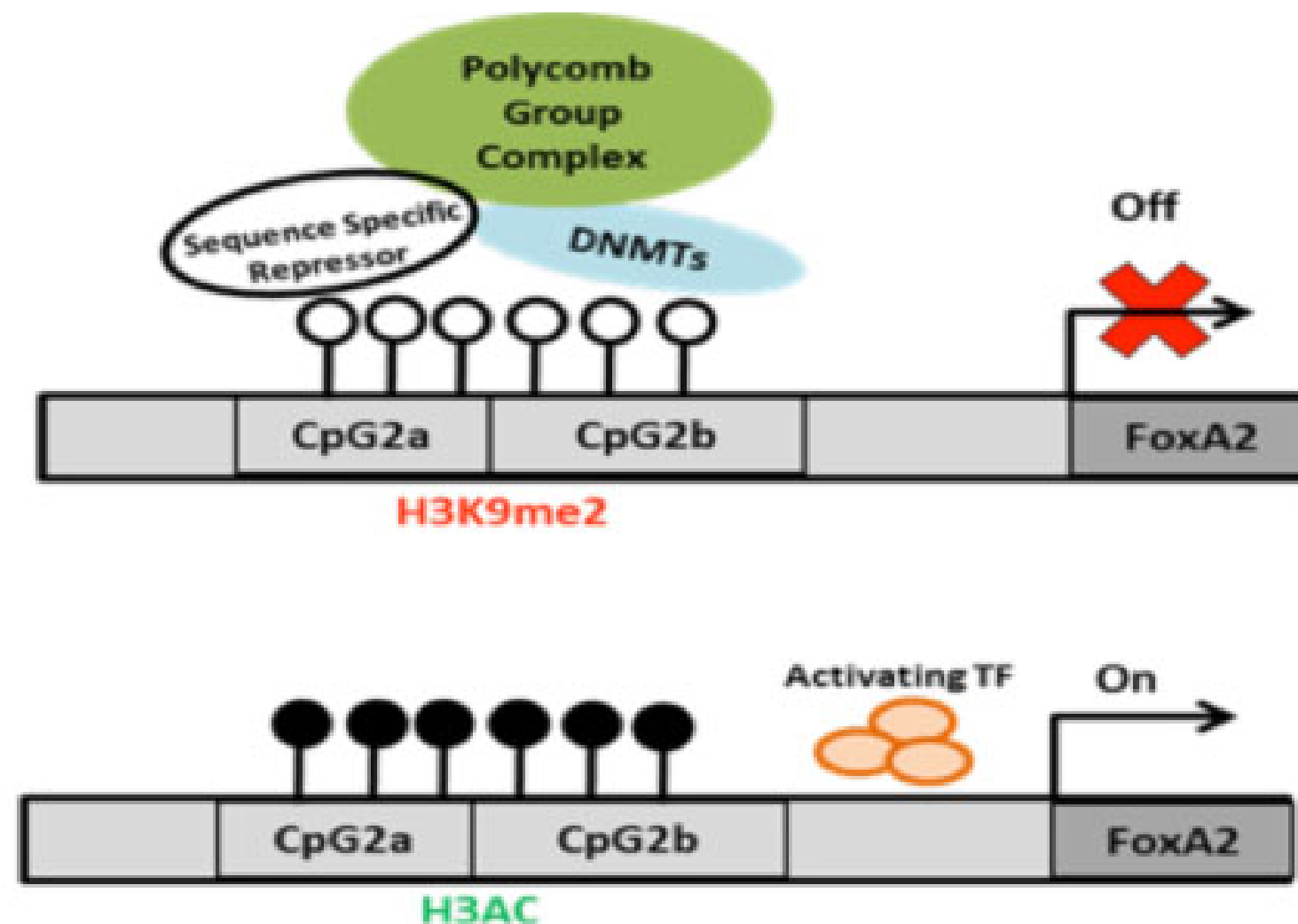
- Induced pluripotent cells: somatic cells that have been reprogrammed back into pluripotent state
- iPSCs enable lineage modeling
- Transcription factors drive fate
- FOXA2 is a key regulator of lineage specification

https://en.wikipedia.org/wiki/Induced_pluripotent_stem_cell

Cerneckis, J., Cai, H., & Shi, Y. Signal Transduction and Targeted Therapy 9, 112 (2024).

Research gap

- The impact of specific FOXA2 mutations is still poorly characterized.



We do not know if it influences:

- FOXA2 structure?
- DNA binding properties?
- Interaction networks?

<https://en.wikipedia.org/wiki/FOXA2>

Sahoo, S. S. *Journal of Clinical Investigation* 132(12): e157574 (2022).

Giri, D. *Human Molecular Genetics* 26(22): 4315–4326 (2017).

Hypothesis

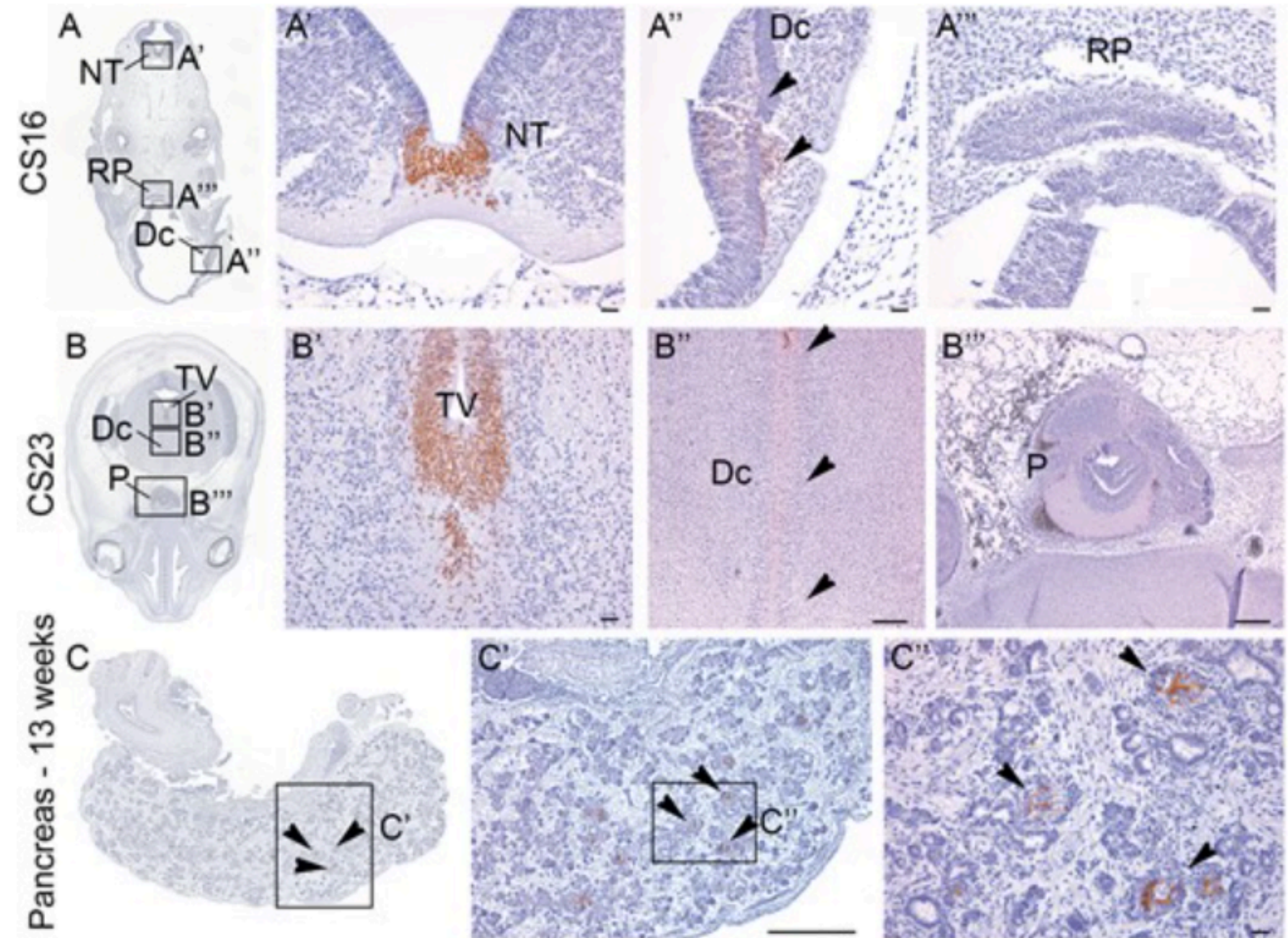
Previous Research

- FOXA2 S169P mutation leads to diseases associated with liver-specific gene and diseases like congenital hypopituitarism (CH).

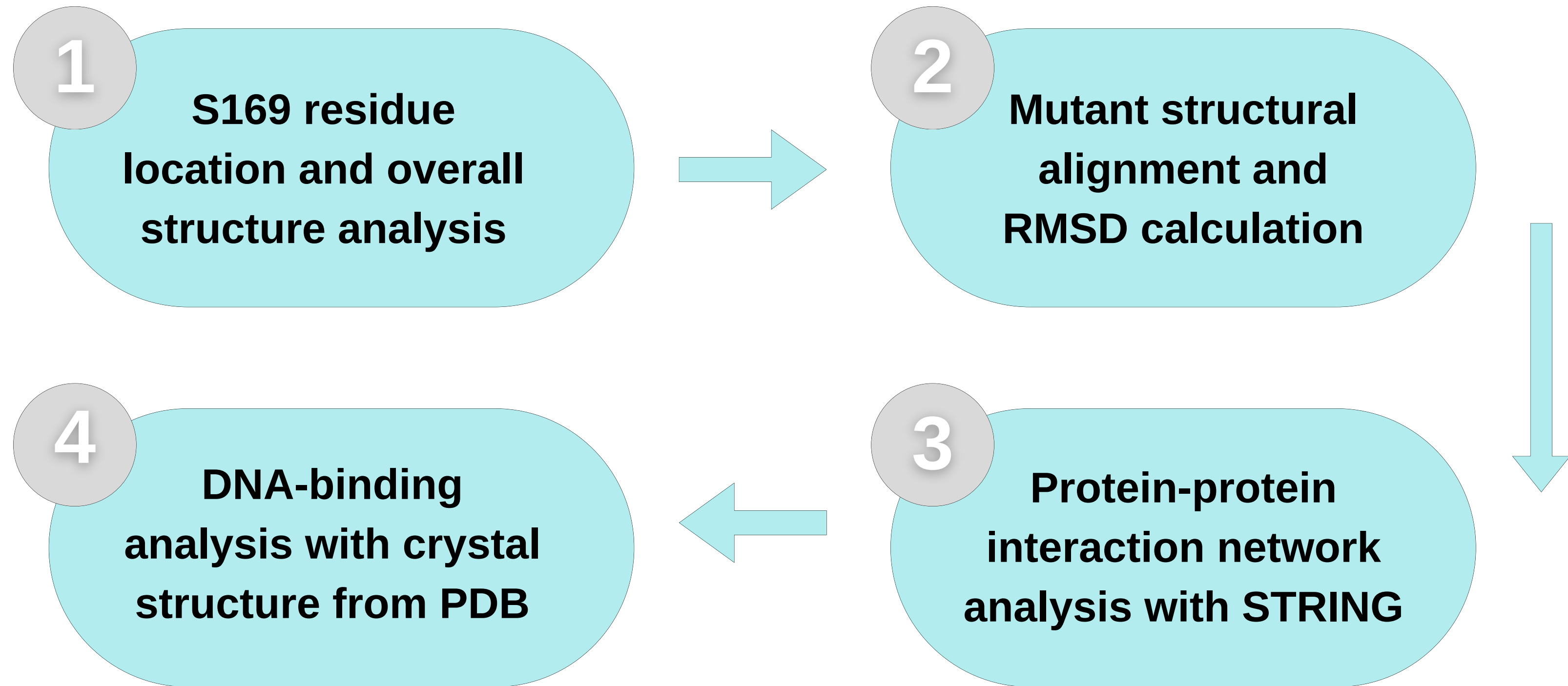
Hypothesis

- perturb protein structure
- interaction properties

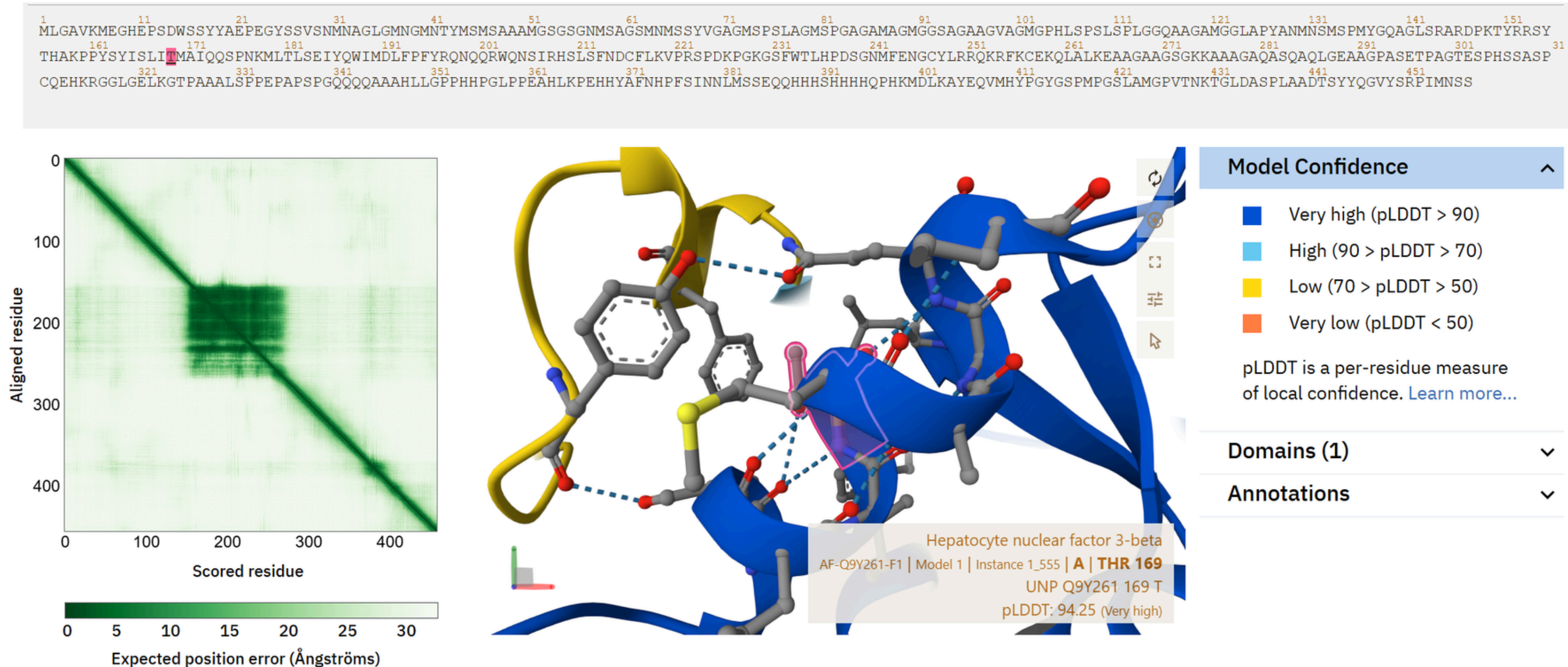
influence regulatory function



Workflow

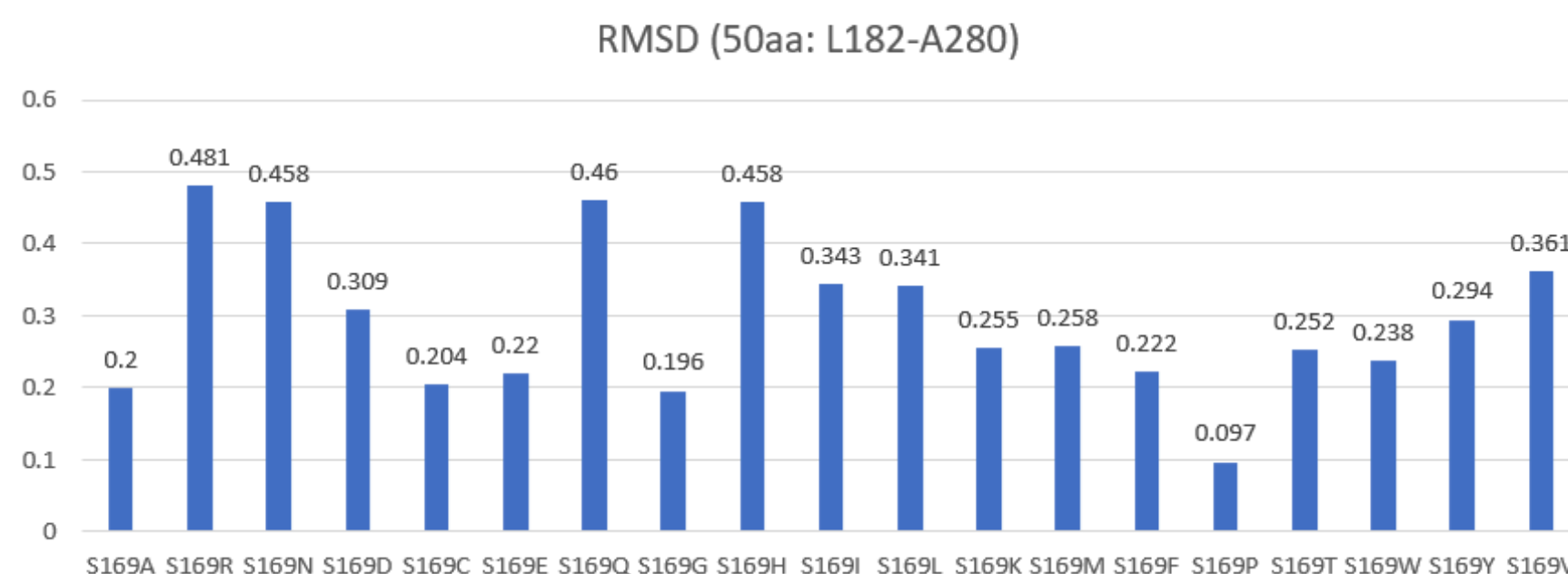
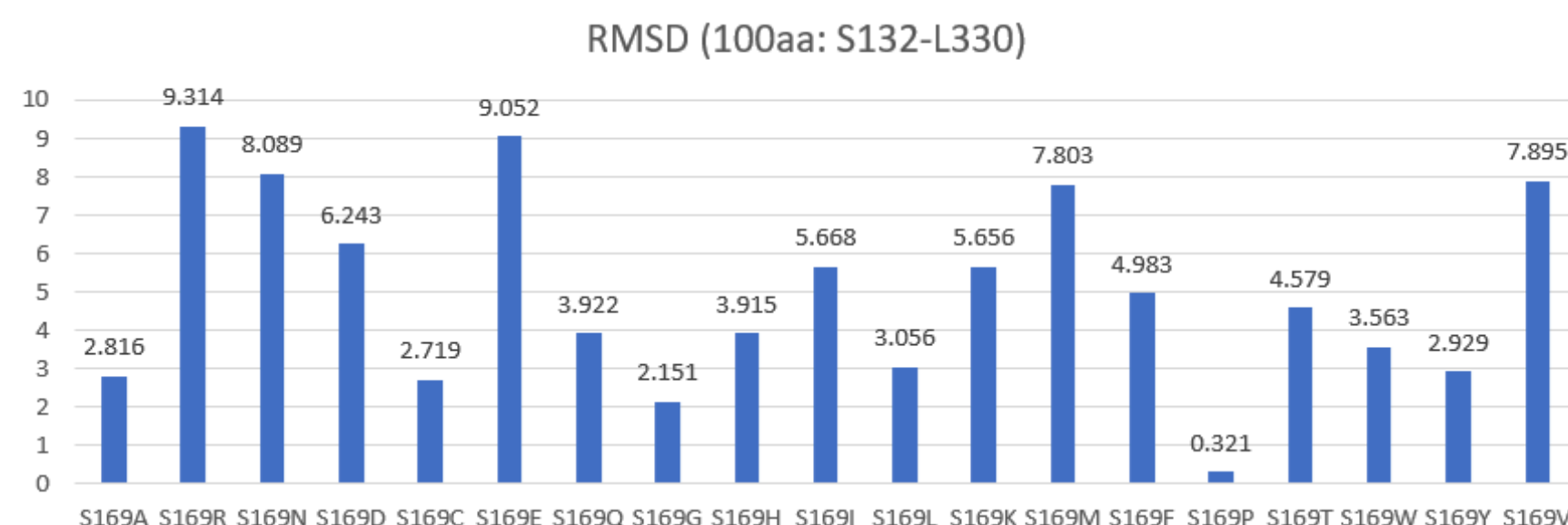


Structural localization

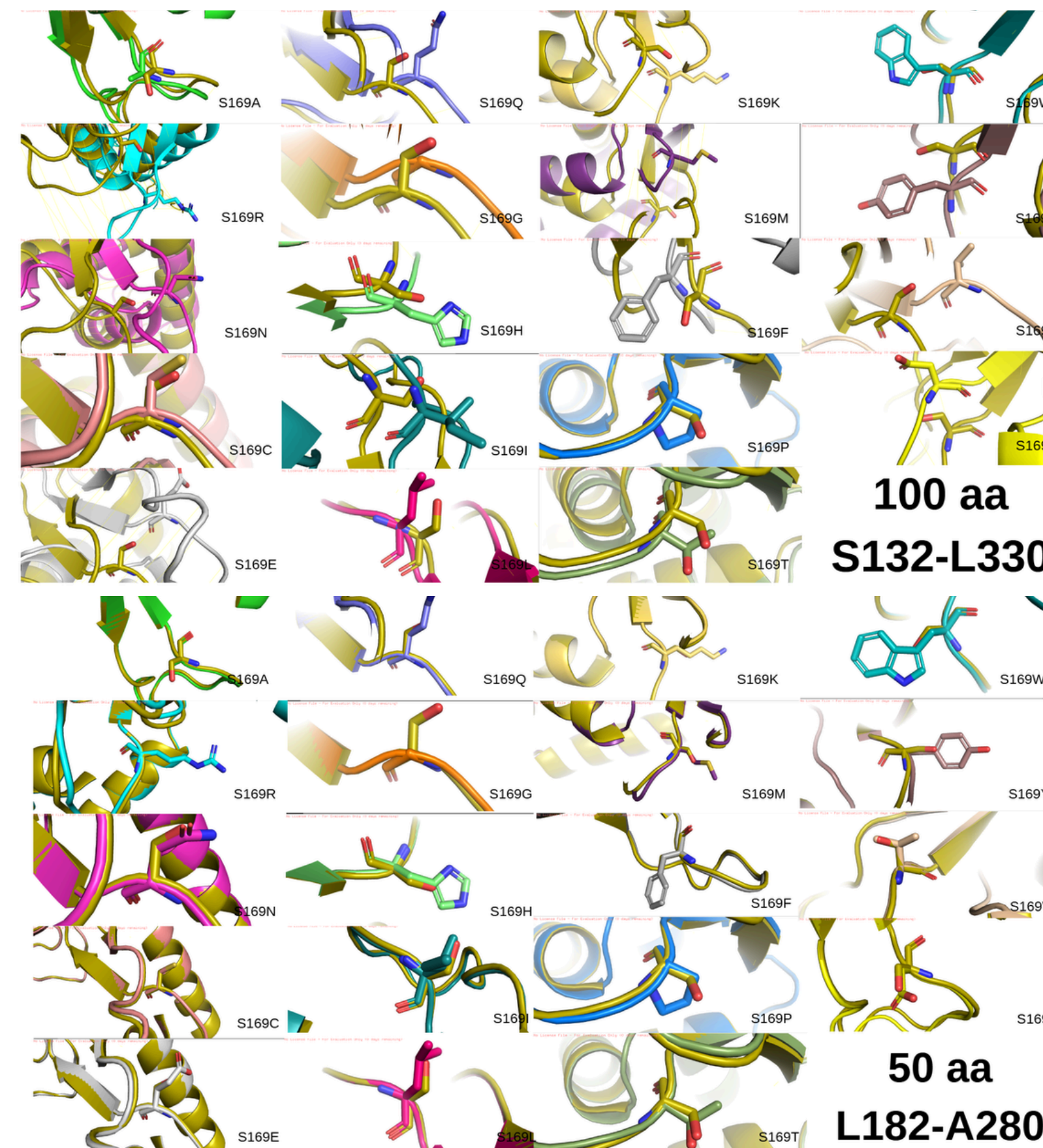


- AlphaFold showed that S169 is located within alpha-helical region
- It lies in a high-confidence segment (high pLDDT)

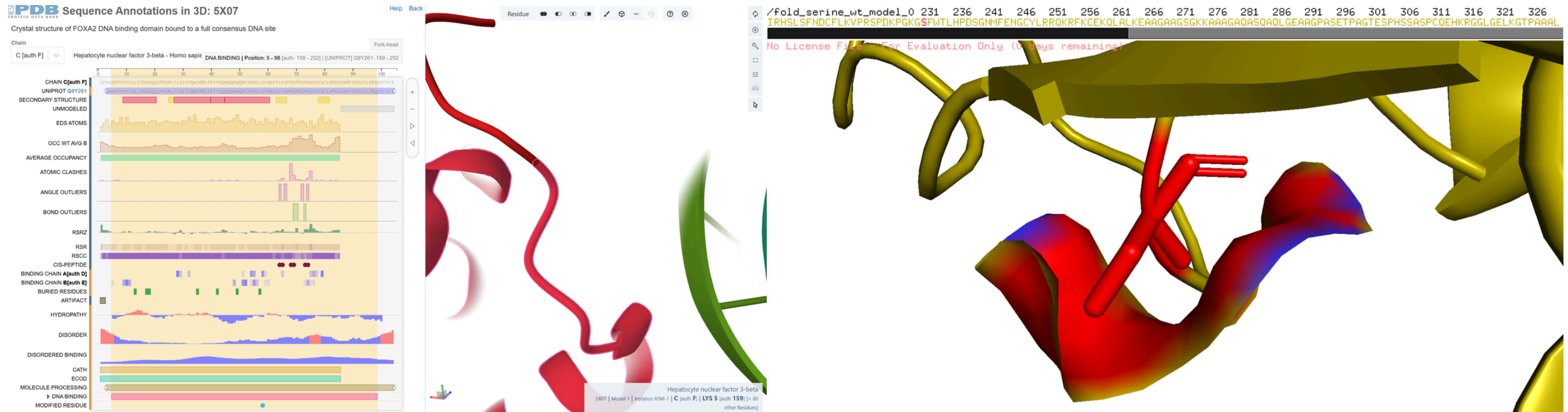
Mutant structure alignment



- The pathogenic potential of S169P, if present, is unlikely to arise from large-scale structural destabilization.



DNA-binding region investigation



- PDB crystal structure: S169 does not directly contact DNA within the binding region.
- S169 is surface-exposed, therefore it is more likely to interact with other proteins.
- **S169 is structurally constrained and surface-exposed but not DNA-contacting.**

Protein-protein network analysis

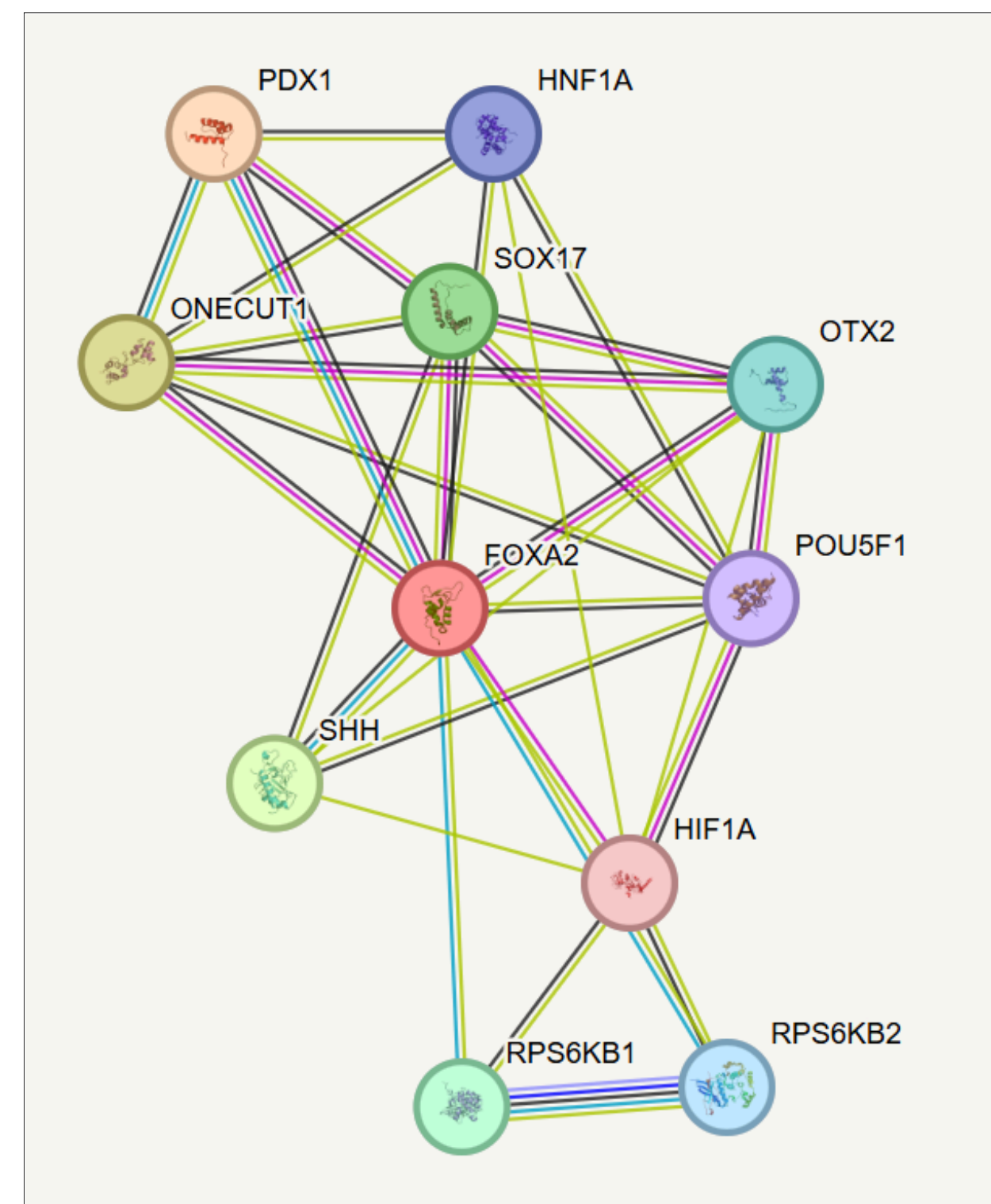
number of nodes: 11
number of edges: 31
average node degree: 5.64
avg. local clustering coefficient: 0.764

expected number of edges: 13
PPI enrichment p-value: 1.19e-05

your network has significantly more interactions than expected (what does that mean?)

- Many interactive proteins are associated with endoderm development and lineage specification

Protein	Function
PDX1	Key transcription factor for pancreatic development and β -cell insulin regulation
HNF1A	Regulates gene expression in liver, pancreas, kidney, controlling glucose metabolism
ONECUT1	Drives liver and pancreatic endocrine cell differentiation
SOX17	Controls endoderm formation and early organ lineage specification
OTX2	Directs early embryonic development and regulates brain formation
POU5F1	Maintains pluripotency and self-renewal in embryonic stem cells
HIF1A	Mediates cellular response to hypoxia by activating survival and metabolic genes
RPS6KB1	Promotes protein synthesis and cell growth through mTOR signaling
RPS6KB2	Enhances protein synthesis and cell proliferation via protein S6 phosphorylation



- FOXA2 sits in a dense developmental regulatory network.**

Network: STRING v12

Biological Interpretation

S169P may subtly modulate interactions with interactive proteins.

- May affect the function of the interactive proteins
- Lead to developmental abnormalities

AlphaFold may ignore variance.

- May ignore variance considering optimizing the overall structure
- Therefore the protein structure may still contribute to consequences

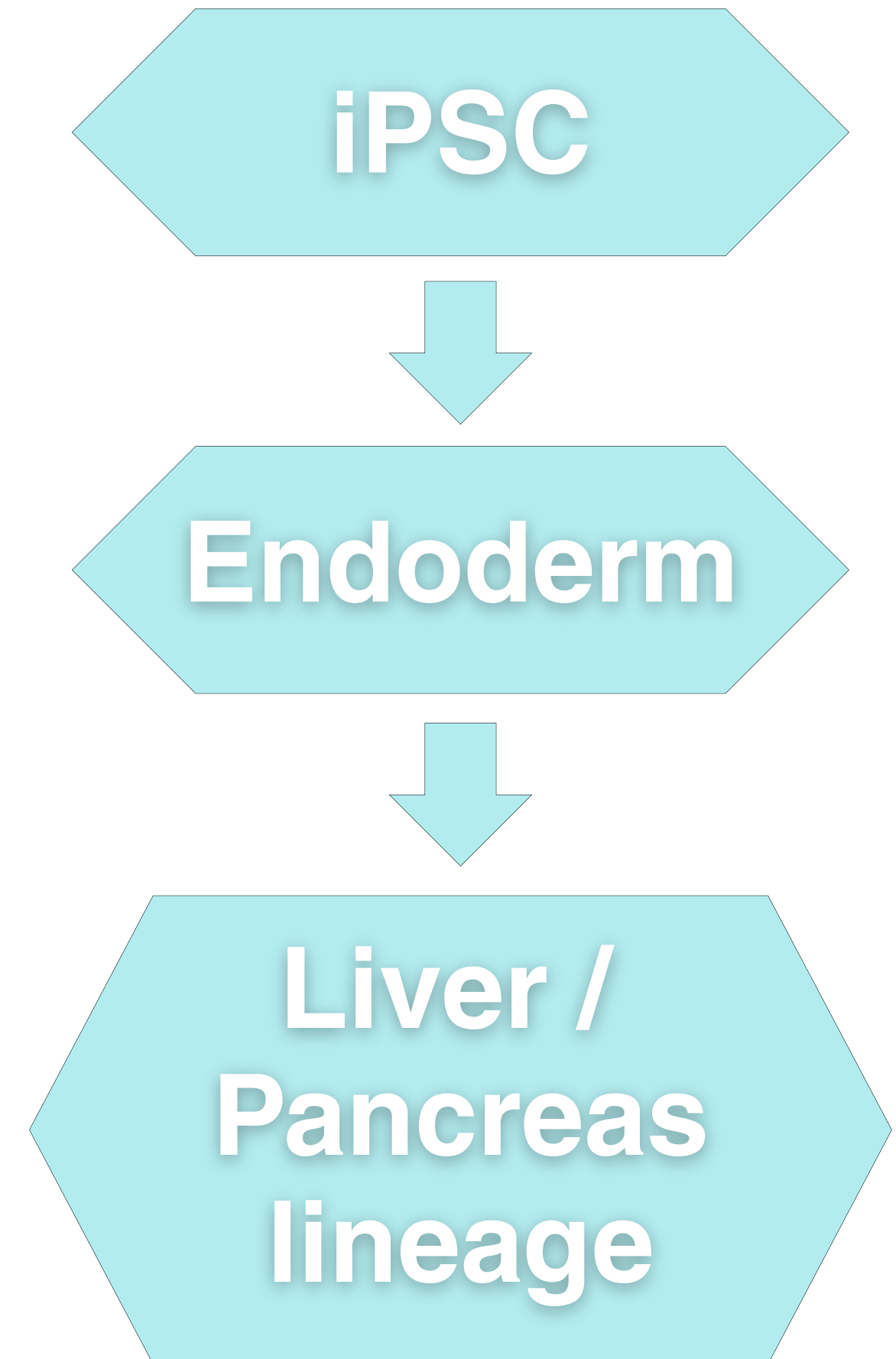
Mutation may influence DNA-binding indirectly

- May make it less possible to release DNA.

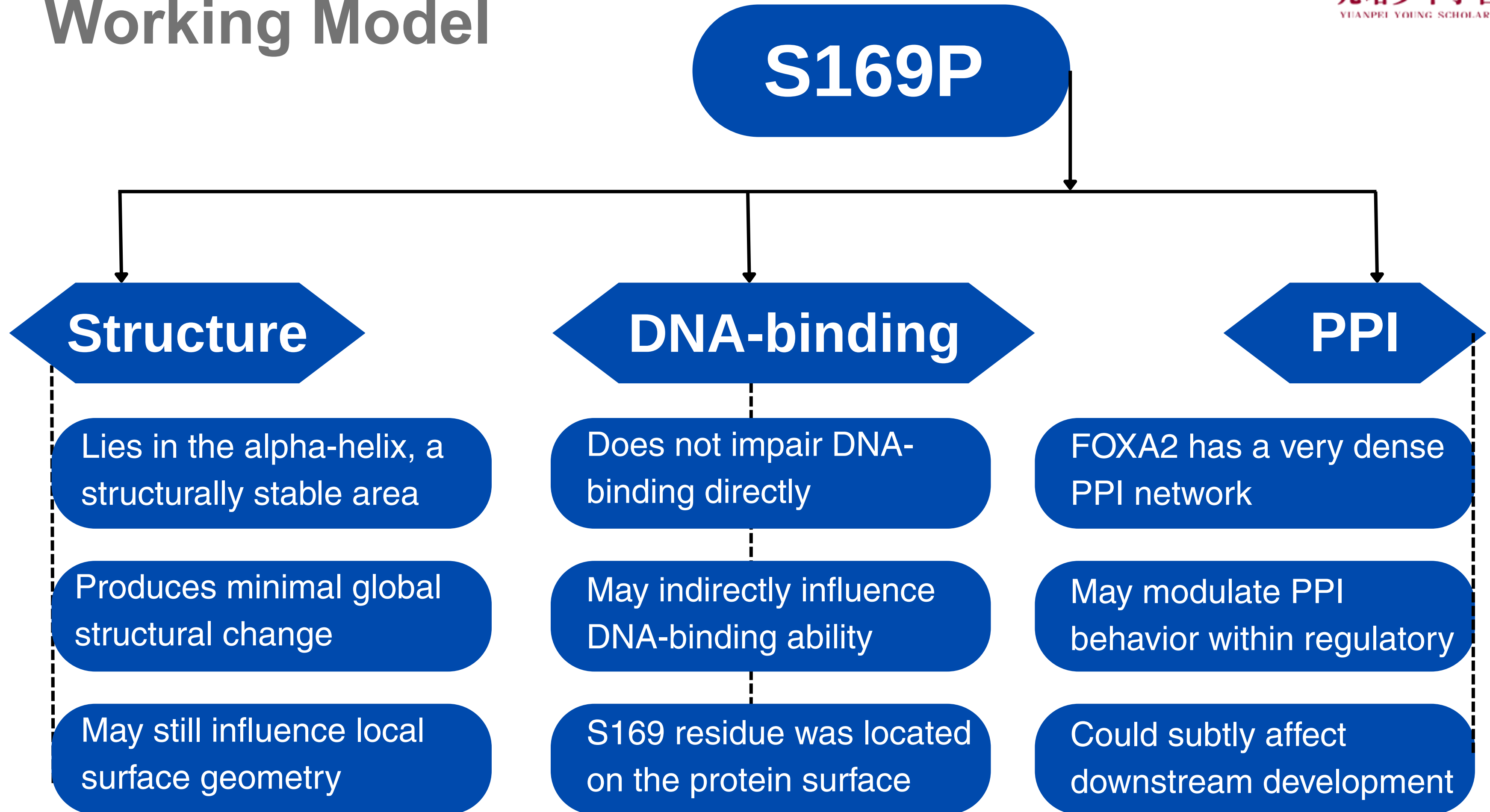
iPSC differentiation implication

- FOXA2 is a key pioneer transcription factor in endoderm specification
- Surface exposure of S169 may influence FOXA2 interaction capacity
- FOXA2 related network is critical for lineage commitment

→ **FOXA2 structural features may contribute to transcriptional control during early lineage specification**



Working Model



Conclusion

- **S169 is located in a structurally constrained alpha-helical region of FOXA2**
- **The S169P substitution causes minimal global structural deviation.**
- **The surface-exposed position suggests a potential role in modulating protein interaction dynamics rather than DNA binding.**

| Future work

- **Test if AlphaFold accurately predict the protein structure**
- **Calculate RMSD using the full amino acid sequence**
- **Further investigation on protein-protein interactions**

感谢老师批评与指导

Thank you for your constructive feedback!

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